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AMBROSE ALLI UNIVERSITY, EKPOMA

CSC 308 — Past Questions & Materials

by Renance

Course: CSC 308

Institution: Ambrose Alli University, Ekpoma

Papers in this pack: 1

Levels: 300

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Paper 1 of 1 — CSC 308

Level 300 • First Semester • PastQuestion

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AMBROSE ALLI UNIVERSITY, EKPOMA
FACULTY OF NATURAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION FOR 2017/2018 ACADEMIC SESSION

CSC 308: INTRODUCTION TO OPERATIONS RESEARCH

INSTRUCTION: Answer any FOUR questions

TIME 2 ½ Hours

1. (a) What is Operations Research? (b) List five solution methods for solving problems in Operations Research
- (c) List and explain the stages involved in solving any problem of interest in Operations Research.
- (d) A boat manufacturer produces two types of boats for the sport and camping trade: a family row boat and a sport canoe. The boats are modeled from aluminum by means of a large pressing machine and are finished by hand labour. A row boat requires 50kg of aluminum, 6 minutes of machine time and 3 hours of finishing labour; A canoe requires 30kg of aluminum, 5 minutes of machine time and 5 hours of finishing labour. For the next three months, the Company can commit up to 2000kg of aluminum, 5 hours of machine time and 200 hours of finishing labour for the manufacture of the small boats. The company realizes ₦ 350 profit on the sale of a row boat and ₦ 420 of canoe. Assuming that all boats produced are sold, how many of each type should be manufactured in the next three months in order to maximize profit.

2. (a) State the mathematical model of the transportation problem
- (b) A canned good supplier has two warehouses serving four outlets. The Northern warehouse has 700 cases on hand and the southern warehouse has 900 cases on hand. The transportation cost in naira per case and the requirement for the four outlets are given in the table below:

	Outlets			
	I	II	III	IV
Northern warehouse	4	3.2	6	4
Southern warehouse	9	8	10	8.5
Requirement (cases)	250	400	450	500

Supply

Write the mathematical model of the transportation problem and determine a transportation schedule that minimizes the costs.

3. (a) Explain the concept of unboundedness, degeneracy and redundancy
- (b) Show that given any primal problem, the dual of its dual is again the primal problem.
- (c) Write the dual of the following linear programming problem.

AMBROSE ALLI UNIVERSITY, EKPOMA
 FACULTY OF PHYSICAL SCIENCE
 DEPARTMENT OF COMPUTER SCIENCE
 2020/2021 FIRST SEMESTER EXAMINATION
CSC 308: INTRODUCTION TO OPERATIONS RESEARCH

Instruction: Answer ant for questions

Time 2 ½ hours

- 1 (a) Given the following lectures preparation time tableaux.

Builder	Computer Hardware	Automata theory	Real-Time Computing	Compiler Construction
Nnebe	2	10	9	7
Momodou	15	4	14	8
Aliga	13	14	16	11
Onianwa	4	15	13	9

(b) State the mathematical model of the assignment problem

(c) Assign the four lecturers to the four courses on one-to-one basis so that the assignment will give the minimal preparation time possible.

- 2 (a) State the mathematical model of the transportation problem.

5	4	5	9
2	3	8	16
6	8	8	

(b) Solve the transportation problem to provide a shipping plan which will minimize the total transportation cost from all sources to all destinations

- 3 Consider the following pay off (profit) matrix where the probabilities for the occurrence of the nature states are unknown. Compare the solutions obtained by each of the following criteria: (a) Laplace (b) Maximin (c) Savage (d) Hurwicz assuming $\alpha = 0.5$

	β_1	β_2	β_3	β_4	β_5
α_1	15	10	0	-6	17
α_2	3	14	8	9	2
α_3	1	5	14	20	-3
α_4	7	19	10	2	0

- 4 State the solution involved in the Dual simplex method of solving linear programming problem.

(b) Minimize $Z = 25x_1 + 35x_2$

Subject to $5x_1 + 6x_2 \geq 250$

$8x_1 + 6x_2 \geq 300$

$10x_1 + 20x_2 \geq 700$

$x_1 \geq 0, x_2 \geq 0$

5. Sketch the graph of the linear programming problem.

$$\text{Minimize } z = 3x_1 + 2x_2$$

$$\text{Subject to } 2x_1 + x_2 \geq 8$$

$$2x_1 + x_2 \geq 6$$

$$x_1 + 3x_2 \geq 6$$

$$x_1 + 6x_2 \geq 8$$

$$x_1 \geq 0, x_2 \geq 0$$

From the graph, Identify the: (i) the corner points feasible solution, (ii) feasible region, (iii) two sets of infeasible solution (iv) optimal feasible solution (v) optimal solution

6. (a) State the method of deriving the dual problem from the primal problem

(b) Given the primal problem below, obtain the dual:

$$\text{Maximize } g = 160y_1 + 200y_2$$

$$\text{Subject to } 2y_1 + 4y_2 \leq 40$$

$$18y_1 + 18y_2 \leq 216$$

$$24y_1 + 12y_2 \leq 240$$

$$y_1 \geq 0, y_2 \geq 0$$

(c) Using the requirement space method, obtain the solution of the dual and read off the solution of the primal.

AMBROSE ALLI UNIVERSITY, EKPOMA
 DIRECTORATE OF DIPLOMA & PART TIME PROGRAMMES
 DEPARTMENT OF COMPUTER SCIENCE
 2011/2012 SECOND SEMESTER EXAMINATION
 CSC 308: Introduction to Operations Research
 Instruction: Answer questions one and any other three. Time 2 ½ hours

- 1 (a) State the general form of the activity analysis problem.
 (b) Write the dual of the following linear programming problem:
 Minimize $f(x_1, x_2) = 10x_1 + 4x_2$
 Subject to $3x_1 + 2x_2 \geq 60$
 $7x_1 + 2x_2 \geq 84$
 $4x_1 + 6x_2 \geq 72$
 and $x_1 \geq 0, x_2 \geq 0$
 (d) Using the requirement space method or otherwise obtain the solution to the dual of question 1(b) and read off the values of the original problem.
- 2 (a) Prove that for a given primal problem, the dual of its dual is again the primal problem.
 (b) Write the dual of the primal problem:
 Maximize $z = 4x_1 + 5x_2 + 7x_3 - x_4$
 Subject to $2x_1 - x_2 + 3x_3 + 4x_4 = 10$
 $x_1 + x_2 + x_3 - x_4 \geq 5$
 $x_1 + 2x_2 - 2x_3 + 4x_4 \leq 12$
 $x_1 \geq 0, x_4 \geq 0, x_2, x_3$ are unrestricted in sign.
- 3 (a) State the mathematical model of the transportation problem
 (b) For a given transportation problem, a solution to the problem is given in the tableau below:

5	2	3	6	100
20	60	80	40	
2	7	7	4	80
60	40	80	40	140
60	60	80	120	

Verify whether the solution is optimal. If no, then determine the optimal solution.

- 4 A manufacturing company has recently decided to produce products, A, B and C. The data necessary for this problem are given below.

Department	Size			Capacity per time period
	A	B	C	
Cutting	10.7	5.0	2.0	2,05
Folding	5.4	10.0	4.0	2,10
Packaging	0.7	1.0	2.0	745
Profit contribution per unit	₦10	₦15	₦20	

Formulate the mathematical model and determine a production schedule that maximizes income.

Q4. A company owns two mines: mine A produces 1 tonne of high grade ore, 3 tonnes of medium grade ore and 5 tonnes of low grade ore each day; and mine B produces 2 tonnes of each of the three grades of ore each day. The company needs 80 tonnes of high grade ore, 160 tonnes of medium grade ore and 200 tonnes of low grade ore.

(a). how many days should each be operated if it costs N200 per day to work each mine?

Suppose, in the preceding problem, the production cost at mine A is N250 per day and at mine B is N150 per day.

(b). how should the production schedule be changed? (18mks)

Q5a. A manufacturing outfit requires 10, 12 and 12 units of chemicals A, B and C respectively, for its clients. A liquid product contains 5, 2 and 1 units of A, B and C respectively per jar, and a dry product contains 1, 2 and 4 units of A, B and C respectively per carton. If the liquid product sells for #3.00 per jar and the dry product sells for #2.00 per carton,

(i) Present the initial tableau. (8mks)

(ii) Find the dual of the minimization problem. (4mks)

(iii) Write the matrix of coefficients of the given (primal) problem. (6mks)

Q6. Solve the linear programming problem using the dual simplex method.

$$\text{Minimize } Z = 2X_1 + X_2$$

Subject to:

$$3X_1 + X_2 \geq 3$$

$$4X_1 + 3X_2 \geq 6$$

$$X_1 + 2X_2 \leq 3$$

$$X_1, X_2 \geq 0 \text{ (18mks)}$$

5. (a) State the mathematical model of the assignment problem and describe an efficient algorithm for solving same.

(b) Given the following class preparation time tableau. Assign the four lecturers to the four courses indicated on a one-to-one basis so that the assignment will give the minimum total preparation time possible.

Lecturers	Linear programming	Queueing theory	Optimization theory	Probability distribution
Omosigho	2	10	9	7
Okonta	15	4	14	8
Ofozu	13	14	16	11
Omojola	4	15	13	9

6. Consider the following problem

$$\text{Maximize } 3x_1 + 2x_2 + x_3$$

Subject to

$$3x_1 + a_1x_2 + 2x_3 = b_1$$

$$3x_1 + a_2x_2 + x_3 = b_2$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$$

where a_1, a_2, b_1, b_2 are constants. For specific values of a_1, a_2, b_1, b_2 the optimal tableau is:

	x_1	x_2	x_3	x_4	x_5	b
x_2	0	a_2	0	1	-1	1
x_1	1	a_1	1	-1/3	2/3	2
	0	2	0	d	e	f

where a_1, a_2, d, e, f are constants.

Determine

(a) The values of a_1, a_2, b_1, b_2 that give the optimal solution.

(b) The values of a_3, a_4, d, e, f .

$$2y_1 + y_2 + y_3 = 9$$

$$-y_1 + y_2 + 2y_3 = 5$$

$$3y_1 + y_2 + y_3 = 7$$

$$4y_1 - y_2 + 4y_3 = 1$$

$$y_1 \geq 0, y_4 \geq 0 \text{ and } y_2, y_3 \text{ are unrestricted}$$

AMBROSE ALLI UNIVERSITY, EKPOMA
FACULTY OF NATURAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE
2013/2014 FIRST SEMESTER DEGREE EXAMINATION
CSC 308 - OPERATIONS RESEARCH

INSTRUCTION: Answer questions one and any other three. Time: 2 ¼ hrs

- Define the term Operations Research. (2mks)
- Outline the contribution of Operations Research to real life study. (3mks)
- Enumerate the usual phases of an operations research study. (6mks)
- (i) List the basic sets of elements that defines a mathematical problem. (3mks)
 (ii) Why do we construct a mathematical model? (1mk)
 (iii) Mention one pitfall of mathematical model. (1mk)
 (iv) What is the crucial step in the formulation of mathematical model? (1mk)
- (i) What is the major phase of any operation research project? (1mk)
- Solve the linear programming problem.
 Maximize $Z = 4X_1 + 3X_2$
 Subject to :

$$X_1 + 3.5X_2 \leq 9$$

$$2X_1 + X_2 \leq 8$$

$$X_1 + X_2 \leq 6$$

$$X_1, X_2 \geq 0$$
 - By graphical method. (5mks)
 - Indicate the feasible region. (1mk)
 - Interpret your result. (2mks)

- Define (i) a queue (ii) a queueing system. (2mks)
 - (i) What is the basic characterization of a queue? (1mk)
 (ii) Outline the basic characteristics of a queueing system. (6mks)
 - Trucks are known to arrive at a dock in a Poisson manner at the rate of 8 per hour. Service time distribution is approximated exponentially with an average rate of 5 minutes. The cost of waiting is #10.00 per hour while the cost of servicing is #4.00 per unit.
 - calculate the average number of trucks in the system. (3mks)
 - calculate the average time a truck spends in the queue. (3mks)
 - calculate the average queue length for the trucks. (3mks)
- Define (i) a feasible solution (ii) an optimal solution. (2mks each)
 - Outline the algorithm of the Simplex method. (4mks)
 - Solve the linear programming problem completely using the Simplex method.
 Maximize $Z = 3X_1 + 9X_2$
 Subject to:

$$x_1 + 4x_2 \leq 8$$

$$x_1 + 2x_2 \leq 4$$

$$x_1, x_2 \geq 0. (10mks)$$

AMBROSE ALLI UNIVERSITY, EKPOMA
FACULTY OF ENVIRONMENTAL STUDIES
DEPARTMENT OF BUILDING

First Semester 2016/2017 B.Sc. Degree Examinations

Course Title: Introduction to Operations Research

Course Code: BLD 402

Instruction: Answer any Four (4) Questions

Time: 2 Hours

- 1(a) What is Operations Research?
 (b) State the processes required by an Operations Researcher to accomplish a task
 (c) State the general form of the diet problem
 (d) A Landlord wants to paint his house. It was to be a one-coast job; to satisfy the requirement the paint must possess a viscosity of at least 200 centipoises. Another requirement is that for a desired level of brilliance, there must be at least 14g of chemical ingredient Y in each gallon of the paint. In addition, for a desired degree of durability, at least 30g of another chemical Z must be present in each gallon of the paint. There are two kinds of paint (I and II) available to him. Type I cost \$6 and type II cost \$4 per gallon. The specifications are given.

	Paint I (per gallon)	Paint II (per gallon)
Viscosity (cp)	400	100
Chemical Y(gm)	20	70
Chemical Z(gm)	20	60

The Landlord decides to blend I and II in order to meet the three requirements at a minimum cost.

- (ii) Formulate the mathematical model (ii) Determine the quantity of type I and type II paints that should be used in each gallon of the blend (iii) What is the minimum cost of the blend?
- 2(a) What is a linear programming problem?
 (b) Prove that the dual of the dual of a primal problem is the primal problem.
 (c) Given the linear programming problem:
- $$\begin{aligned} \text{Minimize } z &= 12x_1 + 42x_2 \\ \text{Subject to } &x_1 + 2x_2 \geq 3 \\ &x_1 + 4x_2 \geq 4 \\ &3x_1 + x_2 \geq 3 \\ &x_1 \geq 0, x_2 \geq 0 \end{aligned}$$
- (iii) Obtain the dual of the problem.
 (iv) Determine the solution of the dual using the requirement space method. Hence, read off the solution of the primal problem.

AMBROSE ALLI UNIVERSITY, EKPOMA
DIRECTORATE OF PART TIME PROGRAMME
DEPARTMENT OF COMPUTER SCIENCE
2008/2009 SECOND SEMESTER EXAMINATION
CSC 308: INTRODUCTION TO OPERATIONS RESEARCH
Time 2 ½ hours

Instructions: Answer any four questions

1. (a) Explain the concept of a linear program
(b) State the general form of the activity analysis problem
(c) A manufacturing company has arbitrarily decided to produce products, A, B, and C. The data necessary for this problem are given below

Department	Size			Capacity per time period
	A	B	C	
Cutting	10.7	5.0	2.0	2,705
Folding	5.4	10.0	4.0	2,210
Packaging	0.7	1.0	2.0	445
Profit contribution per unit	N10	N15	N20	

- i) Write the problem in a tableau; $z = 10x_1 + 15x_2 + 20x_3$
 ii) Identify the appropriate variables; x_1, x_2, x_3
 iii) Formulate the appropriate objective function; $z = 10x_1 + 15x_2 + 20x_3$
 iv) Formulate the constraint set, and
 v) Set up the appropriate linear program
2. (a) State the mathematical form of a linear programming problem.
 (b) Define the following: i) a solution, ii) a feasible solution, iii) feasible region and iv) corner point feasible solution.
 (c) Using the solution space method, maximize $z = 3x_1 + 5x_2$
 subject to
 $x_1 + 4x_2 \leq 24$
 $x_1 + x_2 \leq 9$
 $3x_1 + x_2 \leq 21$
 $x_1, x_2 \geq 0$

3. (a) Outline the steps involved in the simplex method for solving maximization problem and illustrate same with a diagram
 (b) Suppose y_1 and y_2 are the variables in the optimal basis of the linear program below, obtain all the elements of the optimal basis without using the simplex method.
 Maximize $60y_1 + 84y_2 + 72y_3$
 Subject to $3y_1 + 7y_2 + 3y_3 \leq 10$
 $2y_1 + 2y_2 + 6y_3 \leq 4$
 $y_1 \geq 0, y_2 \geq 0, y_3 \geq 0$